Shivendra Agrawal

Curriculum Vitae

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Summary - I am a 2nd year CS graduate starting PhD from Fall 2019 at CU Boulder. Prior to this, I have experience at Noodle.ai and Bosch as a data scientist and as a co-founder of a start-up. I am interested in Computer Vision, 3D Scene Understanding, Machine Learning, and Reinforcement Learning. I am particularly interested in the application of these in Robotics.

Education

2017–2019 **University of Colorado, Boulder**, *Masters of Science*, *Computer Science*.

- GPA: 4.0/4.0 (Scored highest in ML and Algorithms and 2nd highest in VR)
- Teaching Assistant - Awarded **outstanding TA** for **Algorithms** (2017-2018, 2018-2019)

2009–2014 **Indian Institute of Technology, Kharagpur**, *Integrated M.Sc.*, *Mathematics Computing*.

Projects

Feb 2019 - ongoing **Guide dog robot for visually impaired**, Prof. Brad Hayes, CU Boulder.
We are trying to emulate a guide dog in indoor scenario. We propose a solution based on mobile robot platform. The robot would provide navigation assistance and scene explanation using a physical lease feedback and natural language interface. The goal is to help the visually impaired reach their goal faster and safer.

Feb 2019 - ongoing **Visual memory with sparse demonstrations**, *Prof. Chris Heckman*, CU Boulder.
We are trying to learn the path information using visual cues so that the robot can follow the path or do homing without the 3D map of the environment and with very less demonstrations of the path.

August 2018 - December 2018 **An ARKit app to help people with learning disabilities**, *Virtual Reality*, CU Boulder.
We made an IOS app with a python ML backend that parsed text from images and represented it in a more intuitive form that enabled learning. We also created a text editor which would help a person with dyslexia to just focus on some text.
My role - I designed the python backend, the IOS app, the text editor, came up with a way to translate from 2D text image coordinates to 3D world coordinates and helped conceptualize the product.(Link to project)

April 2018 – May 2018 **Logo Generation using GAN**, *Human Centered Machine Learning*, CU Boulder.
We (2 membered team) created a GAN based Logo generator to help generate original and unique initial logo designs. Tried out various variations like WGAN and ACGAN.
My role - Introduced a new way for labeling training data based on the color of the logo to help train ACGAN, developed the Web app that generates logos based on various criteria. (Link to report)

May 2018 **Twitter Trend Analysis**, *Big-Data Architecture*, CU Boulder.
We (5 membered team) developed a scalable web app using Kafka, Cassandra, and Spark which could classify upto 6k tweets per minute in real-time. Additionally, each trend was also classified using majority classification of it's tweets and similar trends were identified using Doc2Vec similarty metric.
My role - project conceptualization, trend similarity, leaflet and front-end development.(Video demo)

Jan 2018 **Grading Tool**, *Teaching Assistant*, CU Boulder.
Built Selenium based browser automation tool to automatically create prettified report cards from Google Sheets and post to nearly 300 students' Moodle or Canvas accounts.

Oct 2017 – Dec 2017 **Image Depth Map Estimation**, *Machine Learning*, CU Boulder.
Compared various CNN based architectures including transfer learning, residual layers and fully convolutional networks using Keras. Demonstrated an application of the predicted depth map to create *"Portrait Mode"*. (Link to report)

Oct 2017 – Dec 2017 **Motion Planning with RRT**, *Algorithms*, CU Boulder.
Created a Javascript based GUI implementation of RRT based algorithm for path finding. Compared this with vanilla A* varying multiple settings including sampling percentage, complexity of obstacles and tree expansion distance. (Link to report)

Nov 2015 – Feb 2016 **Splatter - Food recommender**, *Self*.
Co-created an android app based system for food recommendations and discovery based on preference, vicinity and friends. Designed the tag suggestion system, feed generation system (time-distance decay model for a dynamic feed) and notification system. (Link to App - No longer active)

Papers

September 2018 **Improving Human-Robot Collaboration through Autonomous Explanation-based Reward Coaching**, *Best paper nominee, HRI 2019*, CU Boulder.
We introduced a new framework to infer the human collaborators' mental view of the optimum state, predict future failure based on the inferred human mental model and provide interpretable feedbacks to help the human in a shared task. (Link to initial version of the paper)

Work Experience

Sep 2016 – May 2017 **Associate Data Scientist**, Noodle.ai, Bangalore, India.

- ***Exploratory Data Analysis (EDA) tool*** - Developed a Django app to perform preliminary data analysis. The EDA tool automatically munges data to plot the features, performs statistical tests, find interesting inferences which may be overlooked if there are a lot of features/data and summarizes the result. The tool used parallelization whenever the operations don't need to be sequential.
- ***Demand forecast demo tool*** - Developed a demo tool which uses dummy data and shows micro-demand (store-level) predictions focusing FMCG companies. This tool was used a multiple times and proved to be more effective than a slideshow.
- ***Churn analysis*** - Created a model for churn prediction for a fashion e-commerce brand. Developed reusable, automated pipeline to perform all initial univariate/bivariate analysis and plotting.

Mar 2016 – Aug 2016 **Co-Founder**, Progyrus, Bangalore, India.

- Progyrus provided (Uber-like) large scaled human powered translations to help brands go vernacular. We worked with 6 clients to pilot regional content with 50 translators on the platform
- As one of the co-founders I contributed in conceptualizing the idea and launched the MVP product. Designed the order management and task management/allotment systems.
  Developed dashboards using AngularJS. Also used Django REST framework + PostgreSQL and developed APIs to automate the translation workflow and integration at client side.

June 2014 – Feb 2016 **Senior Engineer (Analytics)**, Robert Bosch GmbH, Bangalore, India.

- **Prototype optimization for an automobile manufacturer** - Developed code to computationally form feasible prototypes (satisfying all boolean formulas) to reduce the number of actual test car productions. Devised and parallelized the algorithms that modeled the problem into a maximal clique finding problem.
- **Knowledge base for automobile domain** - Worked with a team to co-develop a system to gather and exploit unstructured knowledge from automobile specific publicly available domains/forums. The ontology aimed at assisting mechanic identify the symptom, problem and solution. Used dependency parsing, word2vec, fuzzy similarity and regular expressions to build the system.
- **Analysis for delay** - Developed a Python based GUI (tkinter) tool for the upper management to take data of purchase requests at regular interval and generate all insights into the delays and reveal all the bottlenecks and recurring issues.
- **Ticket Analysis** - Analyzed texts in complain tickets and came up with a novel hierarchical clustering method using fuzzy similarity to identify common problems and sub-problems. Made D3.js visualization for easier consumption of the insights.

Competitions

2015 **D4D Data Challenge**, Orange Telecom.
Developed a system using markov clustering to analyze population movement and commute clusters for Senegal using cell phone data. We presented a poster on our analysis at NetMob 2015 in MIT, Cambridge.

2013 **HackU**, Yahoo, IIT Kharagpur, India.
Developed a content editable, version controlled news article system using GitHubPy API & flask backend. It exploited Git's features and provided readers control to edit articles to some extent, reducing news bias.

2012 **Hackathon**, Microsoft, IIT Kharagpur, India.
Developed a Windows phone 7.5 application in C# which aimed at helping the user find their geographically nearest friend with the required blood group in case of emergency.

2011 **GE Edison Challenge**, General Electrics, Bangalore, India.
Designed futuristic energy-efficient green solutions for high-rise complexes and developed physical working models for water usage optimization. Selected in the Top 10 teams (5 membered) from all over India.

Skills & Abilities

Languages	Python, Javascript, R, C++
Frameworks	AngularJS, Django REST, Flask
Libraries	Keras, Scikit-Learn, Selenium, Elasticsearch, Tensorflow
Databases	MongoDB, PostgreSQL

Courses at CU Boulder

ML	ML, NLP, Human-Centered ML, Probabilistic Models
Theory	Algorithms
Others	Virtual Reality & Augmented Reality, Big-Data Architecture

Interests

Teaching	I enjoy teaching and I have taught study groups and under-privileged children apart from my TA experience
Food	I like exploring various cuisines and currently have 4 million total views of mainly food on Google Maps.
Reading	I enjoy reading on the internet and spend some time daily on Hacker News, Reddit and Quora.
Pool	I am a beginner but I am enjoying the learning process.